

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Transformer Mechanics Worksheet

Course Code:	CSA6502	Course Title:	Generative AI and Large Scale Model
CO Assessed:	CO1-limitation (BL3)	Assessment:	Assessment Tool 3– Transformer Mechanics Worksheet
Weightage:	10% of CIA	Total Marks:	20
CO1 Statement:	Explain the fundamentals of Artificial Intelligence, Generative AI, Foundation Models, Transformer architecture, tokenization, embeddings, and Large Language Models.		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

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Question 1: Tokenization

(a) Split "DISAGREEMENT" into 3 sub-word pieces

DIS + AGREE + MENT

Justification: This split separates the word into recognizable, reusable morphological units that a sub-word tokenizer (e.g., BPE or WordPiece) is likely to have learned from other words in its training data:

- "DIS" — a common negating prefix, also seen in "dislike," "disorder," "distrust."
- "AGREE" — the core root word, high-frequency on its own ("agree," "agreed," "agreeing").
- "MENT" — a common noun-forming suffix, also seen in "argument," "government," "enjoyment". Because each piece independently appears in many other common words, the tokenizer's vocabulary is very likely to contain all three as learned merges, even if "DISAGREEMENT" itself never appeared as a whole token during training.

(b) Why sub-word tokenization helps

Sub-word tokenization lets the model represent any word — including rare or unseen ones — as a combination of smaller, frequently-seen pieces instead of needing every whole word in its vocabulary. Since the model has already learned the meaning and usage patterns of pieces like "DIS," "AGREE," and "MENT" from many other words, it can compose an understanding of "DISAGREEMENT" without ever having seen that exact word, which avoids the out-of-vocabulary problem that whole-word tokenizers face.

Question2:Embedding

(a) Which pair has higher cosine similarity?

"dog" and "puppy" would have the HIGHER cosine similarity.

These two words are semantically related — they refer to the same animal (an adult dog vs. a young dog) and appear in very similar contexts (e.g., "walk the __", "feed the __", "pet the __"). "dog" and "truck" are unrelated in meaning and typically appear in very different contexts, so their embedding vectors would point in more different directions, giving a lower cosine similarity. Embedding models learn similarity largely from how words are used in surrounding context, so words that co-occur with similar neighboring words end up closer together in the vector space.

(b) What an embedding represents vs. a one-hot vector

An embedding is a dense vector of real numbers (typically hundreds of dimensions) that encodes a word's meaning based on the contexts it tends to appear in — words with similar meanings or usage patterns end up close together in this vector space, and the geometric relationships between vectors (distance, direction) can even capture semantic relationships like analogies. A one-hot vector, in contrast, is a sparse vector as long as the entire vocabulary, with a single 1 marking that word's index and 0s everywhere else; it treats every word as equally different from every other word and carries no information about meaning or similarity at all.

Question3:Self attention Worked Example

"The city council refused the protesters a permit because they feared violence."

(a) "They" refers to:

☒ City council

(b) Why self-attention (not word order) is needed

A simple word-order rule (like "pronouns refer to the nearest preceding noun phrase") would incorrectly point "they" to "protesters," since that phrase is closer in the sentence. Resolving the reference correctly requires understanding the meaning of the whole sentence: it is the council's fear of violence that logically explains why they refused a permit, which is a semantic/contextual judgment rather than a positional one. Self-attention lets "they" directly attend to and weigh information from every other word in the sentence — regardless of distance — so the model can use cues like "refused" and "feared violence" to infer that "city council" is the more plausible antecedent.

[3.5 marks]

Question4:Query, key, value

In self-attention, each input token is projected into three separate vectors — Query, Key, and Value — each capturing a different role in the attention computation:

- Query (Q): represents what the current token is "looking for" — the kind of information it needs from other tokens in order to update its own representation.

- Key (K): represents what each token "has to offer" — a label or signature that other tokens can match against when searching for relevant information.
- Value (V): represents the actual content or information that gets passed along once a token is deemed relevant — this is what actually gets aggregated into the output.

To combine them, the model computes a similarity score (typically a scaled dot product) between a token's Query vector and every other token's Key vector; these scores are passed through a softmax to produce attention weights that sum to 1. Each token's output representation is then a weighted sum of all the Value vectors, using those attention weights — so tokens whose Keys best match the Query contribute the most to the result.

[5 marks]